MASTERMIND SCHOLARS EDUCATIONAL CENTRE

(MSEC)

"Dream, Believe and Achieve Excellence"

BIOLOGY

THE SKELETAL SYSTEM

Movement and Support in Mammals

200 WASSCE-STANDARD OBJECTIVE QUESTIONS

Covering Part 1 and Part 2 of the Mastermind Notes • Diagram-based Questions • Full Answer Key

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INSTRUCTIONS TO CANDIDATES

1. Each question is followed by four options lettered **A** to **D**. Find the correct option for each question.
2. Answer all 200 questions. Questions 193 to 200 are based on diagrams; study each diagram carefully before answering.
3. Read every stem carefully. Watch for the words **NOT** and **except**.
4. Suggested timing: attempt each section in one sitting (about 50 questions per hour, as in the WASSCE Paper 1).
5. Mark your work with the answer key at the back only after you have finished.

COVERAGE

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SECTION A: MOVEMENT AND TYPES OF SKELETON

1. The movement of a whole organism from one place to another is called

A. irritability
B. excretion
C. locomotion
D. growth

2. Movement in mammals is brought about by the combined action of the

A. skeletal, muscular and nervous systems
B. respiratory, skeletal and circulatory systems
C. skeletal, digestive and nervous systems
D. muscular, excretory and endocrine systems

3. The skeleton of an earthworm is described as

A. an endoskeleton
B. cartilaginous
C. hydrostatic
D. an exoskeleton

4. The exoskeleton of a grasshopper is made mainly of

A. cartilage
B. chitin
C. keratin
D. bone

5. In crustaceans such as the crab, the exoskeleton is further hardened by

A. calcium salts
B. silica
C. collagen
D. keratin

6. The shell of a snail is made mainly of

A. chitin
B. cellulose
C. keratin
D. calcium carbonate

7. Which of the following animals has a skeleton made entirely of cartilage?

A. Frog
B. Shark
C. Tilapia
D. Lizard

8. Moulting is necessary in arthropods because the exoskeleton

A. becomes too soft with age
B. cannot grow with the animal
C. is made of living cells
D. absorbs too much water

9. Which of the following is an advantage of the exoskeleton to terrestrial insects?

A. It is made of living tissue
B. It stores food reserves
C. It prevents desiccation
D. It allows continuous growth

10. An arthropod is most vulnerable to predators

A. during mating
B. soon after moulting
C. when its exoskeleton is fully hardened
D. while feeding

11. Vertebrates can grow larger than arthropods mainly because the endoskeleton

A. grows with the body and is relatively light
B. is made of chitin
C. is covered by a cuticle
D. has no joints

12. In the earthworm, contraction of the longitudinal muscles makes a segment

A. shorter and fatter
B. longer and thinner
C. coil into a ball
D. stiff and straight

13. The structures which prevent an earthworm from slipping backwards during locomotion are the

A. tentacles **B.** cilia
C. parapodia **D.** chaetae

14. The fluid which acts as the skeleton of an earthworm is held in the

A. coelom **B.** gut
C. blood vessels **D.** nephridia

15. Which of the following animals does NOT have a hydrostatic skeleton?

A. Jellyfish **B.** Sea anemone
C. Cockroach **D.** Earthworm

16. Animals with hydrostatic skeletons are mainly restricted to moist or aquatic habitats because

A. they cannot breathe on land
B. water is needed to maintain the skeletal fluid
C. they lack chaetae
D. their muscles need light

17. Which of the following is true of the endoskeleton but NOT of the exoskeleton?

A. It is made of non-living material
B. It must be shed for growth
C. It covers the body surface
D. Muscles are attached to its outer surface

18. The skeleton performs all the following functions except

A. giving shape to the body
B. protection of delicate organs
C. digestion of food
D. providing surfaces for muscle attachment

19. The human penis is described as a hydrostatic organ because it

A. becomes rigid when its spongy tissue fills with blood under pressure
B. contains cartilage
C. is supported by ligaments
D. contains a bone

20. Which of the following lists contains only supporting materials found in animals?

A. Lignin, cellulose and chitin **B.** Cellulose, cartilage and bone
C. Lignin, chitin and bone **D.** Chitin, bone and cartilage

SECTION B: BONE AND CARTILAGE

21. Bone and cartilage are both classified as

A. nervous tissues **B.** connective tissues
C. epithelial tissues **D.** muscular tissues

22. The hardness of bone is due mainly to

A. collagen **B.** keratin
C. calcium phosphate **D.** chondrin

23. The protein which gives bone some toughness and slight flexibility is

A. keratin **B.** myosin
C. actin **D.** collagen

24. The main mineral crystal found in bone and teeth is

A. sodium chloride **B.** hydroxyapatite
C. magnesium sulphate **D.** calcium oxalate

25. The process by which cartilage is gradually replaced by bone is

A. ecdysis **B.** regeneration
C. resorption **D.** ossification

26. Ossification makes the skeleton

A. softer and more flexible **B.** lighter and more elastic
C. harder but more brittle **D.** free of blood vessels

27. Bone-forming cells which secrete the bone matrix are

A. chondrocytes **B.** osteoblasts
C. osteoclasts **D.** osteocytes

28. Mature bone cells enclosed in the matrix are

A. chondroblasts **B.** osteoclasts
C. osteocytes **D.** osteoblasts

29. Which cells break down old bone and release its minerals into the blood?

A. Chondrocytes **B.** Osteoclasts
C. Osteocytes **D.** Osteoblasts

30. The bones of elderly people fracture more easily than those of children mainly because

A. their bones contain more cartilage
B. their bones are more brittle and lose minerals
C. their bones contain more collagen
D. their bones have more blood vessels

31. The basic structural unit of compact bone is the

A. canaliculus **B.** lamella
C. Haversian system **D.** lacuna

32. Blood vessels and nerves in compact bone are found in the

A. lacunae **B.** lamellae
C. Haversian canal **D.** chondrin

33. The concentric layers of matrix around a Haversian canal are called

A. osteons **B.** canaliculi
C. lacunae **D.** lamellae

34. In bone, the osteocytes are found in spaces called

A. lacunae **B.** lamellae
C. Haversian canals **D.** canaliculi

35. The tiny channels which link the lacunae to one another and to the central canal are

A. trabeculae **B.** sinuses
C. canaliculi **D.** lamellae

36. The matrix of cartilage is called

A. synovial fluid **B.** chondrin
C. hydroxyapatite **D.** ossein

37. Cartilage cells are called

A. fibroblasts **B.** osteocytes
C. chondrocytes **D.** osteoblasts

38. Cartilage repairs slowly after injury because it

A. contains Haversian canals **B.** is made of non-living material
C. has no blood vessels **D.** contains calcium salts

39. Which of the following is true of cartilage?

A. It is avascular
B. It contains Haversian systems
C. It is porous
D. Its matrix contains calcium salts

40. The type of cartilage found in the intervertebral discs is

A. fibrocartilage
B. calcified cartilage
C. hyaline cartilage
D. elastic cartilage

41. The type of cartilage which covers the ends of bones at joints is

A. elastic cartilage
B. calcified cartilage
C. fibrocartilage
D. hyaline cartilage

42. The pinna and the epiglottis are supported by

A. elastic cartilage
B. bone
C. hyaline cartilage
D. fibrocartilage

43. Which of the following is a similarity between bone and cartilage?

A. Both have blood vessels
B. Both have cells in lacunae
C. Both have Haversian systems
D. Both have calcium deposits

44. Which of the following is found in bone but NOT in cartilage?

A. Collagen fibres
B. Living cells
C. Haversian canals
D. Lacunae

45. The C-shaped rings which keep the trachea open are made of

A. hyaline cartilage
B. elastic cartilage
C. fibrocartilage
D. bone

SECTION C: THE SKULL AND THE VERTEBRAL COLUMN

46. The axial skeleton consists of the

A. limb girdles and vertebral column
B. skull and limbs
C. skull, vertebral column, ribs and sternum
D. limbs and limb girdles

47. Which of the following bones forms part of the axial skeleton?

A. Fibula
B. Atlas
C. Scapula
D. Humerus

48. The appendicular skeleton is made up of the

A. limbs and limb girdles
B. lumbar and thoracic vertebrae
C. ribs and sternum
D. skull and vertebral column

49. The only freely movable bone of the mammalian skull is the

A. lower jaw
B. cranium
C. upper jaw
D. occipital bone

50. The bones of the cranium are joined together by immovable joints called

A. facets
B. symphyses
C. condyles
D. sutures

51. The spinal cord leaves the skull through the

A. orbit
B. obturator foramen
C. neural spine
D. foramen magnum

52. The occipital condyles of the skull articulate with the

A. lower jaw **B.** atlas
C. clavicle **D.** axis

53. The main function of the cranium is to

A. protect the brain **B.** protect the lungs
C. attach the limbs **D.** support the teeth

54. The main function of the intervertebral discs is to

A. attach the ribs
B. produce blood cells
C. absorb shock and prevent friction between vertebrae
D. protect the brain

55. Adjacent vertebrae are held together by

A. sutures **B.** ligaments
C. tendons **D.** synovial fluid

56. The number of cervical vertebrae in most mammals is

A. 5 **B.** 7
C. 12 **D.** 13

57. The total number of vertebrae in the human vertebral column is

A. 26 **B.** 33
C. 45 **D.** 60

58. In man, the sacrum is formed by the fusion of

A. three lumbar vertebrae **B.** five sacral vertebrae
C. four sacral vertebrae **D.** four caudal vertebrae

59. The coccyx is formed from fused

A. lumbar vertebrae **B.** thoracic vertebrae
C. caudal vertebrae **D.** sacral vertebrae

60. All the vertebrae of mammals have the following features except

A. neural spine **B.** neural canal
C. vertebrarterial canals **D.** facets for articulation

61. The part of a vertebra which bears the weight of the body is the

A. centrum **B.** transverse process
C. neural spine **D.** neural arch

62. The neural canal of a vertebra carries the

A. spinal cord **B.** oesophagus
C. vertebral artery **D.** aorta

63. The neural spine and transverse processes of a vertebra mainly provide

A. a passage for blood vessels **B.** surfaces for muscle attachment
C. protection for the spinal cord **D.** articulation with the skull

64. The articulating facets at the anterior end of a vertebra are the

A. metapophyses **B.** postzygapophyses
C. prezygapophyses **D.** anapophyses

65. The joints formed between the zygapophyses of adjacent vertebrae are

A. gliding joints **B.** hinge joints
C. pivot joints **D.** ball and socket joints

66. The vertebrarterial canal allows the passage of the

A. spinal cord
B. vertebral artery
C. pulmonary vein
D. aorta

67. The vertebrarterial canal is characteristic of the

A. lumbar vertebra
B. thoracic vertebra
C. cervical vertebra
D. caudal vertebra

68. Cervical ribs are found on the

A. cervical vertebrae
B. sacrum
C. sternum
D. thoracic vertebrae

69. Which vertebra directly carries the skull?

A. Sacrum
B. Axis
C. Thoracic vertebra
D. Atlas

70. The precise location of the atlas is between the

A. sacrum and the caudal vertebrae
B. last thoracic and first lumbar vertebrae
C. axis and the third cervical vertebra
D. skull and the axis

71. The atlas vertebra lacks a

A. transverse process
B. neural canal
C. vertebrarterial canal
D. centrum

72. The joint between the skull and the atlas allows

A. bending of the trunk
B. rotation of the head
C. no movement
D. nodding of the head

73. The odontoid process is found on the

A. sacral vertebra
B. thoracic vertebra
C. atlas vertebra
D. axis vertebra

74. The significance of the odontoid process is that it

A. allows the spinal cord to join the brain
B. strengthens the vertebra
C. provides a point for pivoting
D. provides surfaces for rib attachment

75. The axis vertebra has no

A. centrum
B. prezygapophyses
C. neural spine
D. postzygapophyses

76. Which of the following pairs of features belongs to the atlas?

A. Absent centrum and wing-like transverse processes
B. Metapophyses and anapophyses
C. Long neural spine and facets for ribs
D. Odontoid process and large neural spine

77. A vertebra with a long neural spine pointing backwards and facets for ribs is likely to be

A. an atlas vertebra
B. a thoracic vertebra
C. a lumbar vertebra
D. an axis vertebra

78. The head (capitulum) of a rib articulates with the

A. clavicle
B. sternum
C. demifacets on the centra of thoracic vertebrae
D. facet on the transverse process

79. The tubercle of a rib articulates with the

A. scapula
B. sternum
C. facet on the transverse process of a thoracic vertebra
D. centrum of a lumbar vertebra

80. Which of the vertebrae has the most developed centrum?

A. Lumbar vertebra **B.** Caudal vertebra
C. Cervical vertebra **D.** Thoracic vertebra

81. The lumbar vertebra of mammals has a well-developed centrum in order to

A. bear the weight of the upper body **B.** protect the spinal cord
C. provide attachment for the ribs **D.** carry the vertebral artery

82. Metapophyses and anapophyses are characteristic of the

A. thoracic vertebra **B.** cervical vertebra
C. sacrum **D.** lumbar vertebra

83. The transverse processes of a lumbar vertebra are

A. short and bear facets for ribs
B. perforated by vertebrarterial canals
C. long and point forwards and downwards
D. absent

84. The sacrum provides a firm attachment for the

A. pectoral girdle **B.** skull
C. pelvic girdle **D.** ribs

85. The first sacral vertebra is fused to the ilium by its

A. neural spine **B.** metapophyses
C. sacral wings **D.** odontoid process

86. Which of the following is the correct order of vertebrae from the neck to the tail?

A. Cervical, lumbar, thoracic, sacral, caudal
B. Cervical, thoracic, lumbar, sacral, caudal
C. Cervical, thoracic, sacral, lumbar, caudal
D. Thoracic, cervical, lumbar, caudal, sacral

SECTION D: THE RIBS AND STERNUM

87. The number of pairs of ribs in man is

A. 7 **B.** 10
C. 12 **D.** 24

88. Ribs which are attached directly to the sternum by their own costal cartilage are called

A. true ribs **B.** false ribs
C. floating ribs **D.** cervical ribs

89. The number of pairs of floating ribs in man is

A. 2 **B.** 3
C. 5 **D.** 7

90. Ribs which are NOT attached to the sternum at all are called

A. floating ribs **B.** costal ribs
C. true ribs **D.** false ribs

91. In man, the 8th, 9th and 10th pairs of ribs are

A. true ribs **B.** false ribs
C. floating ribs **D.** cervical ribs

92. The uppermost part of the sternum, to which the clavicles are joined, is the

A. manubrium **B.** acromion
C. xiphoid process **D.** body of the sternum

93. Ribs are joined to the sternum by

A. synovial fluid **B.** sutures
C. tendons **D.** costal cartilage

94. The muscles between the ribs which help in breathing are the

A. biceps **B.** intercostal muscles
C. triceps **D.** deltoid muscles

SECTION E: THE LIMBS

95. A limb which terminates in five digits is described as

A. bipedal **B.** digitigrade
C. pentadactyl **D.** tetrapod

96. The bone of the upper arm is the

A. ulna **B.** humerus
C. radius **D.** femur

97. The forearm is made up of the

A. carpals and metacarpals **B.** humerus and ulna
C. tibia and fibula **D.** radius and ulna

98. The head of the humerus articulates with the

A. sigmoid notch **B.** acetabulum
C. glenoid cavity **D.** obturator foramen

99. The bicipital groove of the humerus allows the passage of the

A. vertebral artery **B.** spinal cord
C. tendon of the biceps **D.** tendon of the triceps

100. The deltoid ridge on the humerus is for the

A. attachment of the clavicle **B.** articulation with the ulna
C. passage of nerves **D.** attachment of the deltoid muscle

101. The supratrochlear foramen of the humerus of a rabbit allows the passage of

A. the spinal cord **B.** the tendon of the biceps
C. a nerve, an artery and a vein **D.** synovial fluid

102. The trochlea of the humerus articulates with the

A. sigmoid notch of the ulna **B.** head of the radius
C. acetabulum **D.** glenoid cavity

103. The structure which prevents the arm from bending backwards is the

A. deltoid ridge **B.** coronoid process
C. patella **D.** olecranon process

104. The olecranon process is a part of the

A. humerus **B.** ulna
C. scapula **D.** radius

105. When the arm is bent, the coronoid process of the ulna fits into the

A. glenoid cavity **B.** olecranon fossa
C. coronoid fossa **D.** acetabulum

106. The larger of the two bones of the forearm is the

A. tibia **B.** fibula
C. radius **D.** ulna

107. The bone of the forearm which allows the hand to rotate is the

A. ulna **B.** carpal
C. radius **D.** humerus

108. The bones of the wrist are the

A. carpals **B.** metacarpals
C. tarsals **D.** phalanges

109. The longest and strongest bone in the human body is the

A. tibia **B.** femur
C. fibula **D.** humerus

110. The head of the femur fits into the

A. sigmoid notch **B.** acetabulum
C. obturator foramen **D.** glenoid cavity

111. The projections near the head of the femur for the attachment of thigh muscles are the

A. tuberosities **B.** trochanters
C. condyles **D.** trochleae

112. The two knobs at the lower end of the femur which articulate with the tibia are the

A. coronoid processes **B.** tuberosities
C. trochanters **D.** condyles

113. The patella prevents the

A. leg from bending forwards at the knee
B. ankle from rotating
C. hip from dislocating
D. arm from bending backwards

114. The bones of the shank are the

A. radius and ulna **B.** tarsals and metatarsals
C. femur and tibia **D.** tibia and fibula

115. The heel bone, which provides attachment for the Achilles tendon, is the

A. metatarsal **B.** calcaneum
C. fibula **D.** patella

116. The number of metatarsals in the hind limb of a rabbit is

A. 4 **B.** 5
C. 6 **D.** 7

117. In man, the thumb and the big toe each have

A. 1 phalanx **B.** 2 phalanges
C. 3 phalanges **D.** 5 phalanges

118. Which of the following pairs of bones occupy corresponding positions in the fore limb and hind limb?

A. Humerus and femur
B. Scapula and tibia
C. Radius and femur
D. Ulna and patella

119. One of the structural adaptations of mammals for movement on land is the development of

A. pentadactyl limbs
B. a streamlined body
C. hollow bones
D. fused thoracic bones

SECTION F: THE LIMB GIRDLES

120. Bones which attach the limbs to the axial skeleton are called

A. phalanges
B. limb girdles
C. vertebrae
D. ribs

121. Each half of the pectoral girdle consists of the

A. clavicle and sternum
B. scapula, coracoid and clavicle
C. scapula and sternum
D. ilium, ischium and pubis

122. The shoulder blade is the

A. clavicle
B. coracoid
C. sternum
D. scapula

123. The collar bone is the

A. manubrium
B. scapula
C. sternum
D. clavicle

124. The glenoid cavity is found on the

A. ilium
B. sternum
C. scapula
D. humerus

125. The ridge which runs along the outer surface of the scapula is the

A. metacromion process
B. acromion process
C. coracoid process
D. spine

126. The clavicle articulates with the scapula at the

A. metacromion process
B. glenoid cavity
C. coracoid process
D. acromion process

127. The pectoral girdle is attached to the vertebral column by

A. muscles
B. fusion with the sacrum
C. the pubic symphysis
D. a pivot joint

128. The main advantage of the loose attachment of the pectoral girdle is that it

A. supports the weight of the body
B. protects the pelvic organs
C. prevents dislocation of the shoulder
D. allows a wide range of movement of the fore limb

129. Each half of the pelvic girdle is called the

A. scapula
B. coccyx
C. innominate bone
D. sacrum

130. The innominate bone is formed by the fusion of the

A. sacrum and coccyx
B. scapula and clavicle
C. ilium, ischium and pubis
D. femur and tibia

131. The bone which fuses the pelvic girdle to the sacrum is the

A. femur **B.** ischium
C. ilium **D.** pubis

132. The two innominate bones are fused together ventrally at the

A. sacro-iliac joint **B.** acetabulum
C. obturator foramen **D.** pubic symphysis

133. The hole between the ischium and the pubis is the

A. acetabulum **B.** glenoid cavity
C. foramen magnum **D.** obturator foramen

134. The obturator foramen of the pelvic girdle

A. provides attachment for the clavicle
B. carries the spinal cord
C. allows the passage of nerves and blood vessels
D. articulates with the head of the femur

135. The pelvis is formed by the

A. pelvic girdle and the sacrum **B.** pectoral girdle and the sternum
C. ilium and the femur **D.** sacrum and the coccyx

136. The acetabulum is a deep socket because the hip joint must

A. allow the passage of nerves
B. produce synovial fluid
C. allow movement in one plane only
D. bear the weight of the body without dislocating

137. Which of the following is a difference between the pectoral and pelvic girdles?

A. The pelvic girdle is fused to the sacrum but the pectoral girdle is not
B. The pectoral girdle has an obturator foramen
C. Both girdles attach the fore limbs
D. The glenoid cavity is deeper than the acetabulum

138. In females, the wide pelvis

A. produces the ova **B.** protects the lungs
C. produces red blood cells only **D.** forms the birth canal

SECTION G: JOINTS

139. A point where two or more bones meet is a

A. ligament **B.** girdle
C. joint **D.** tendon

140. The joint between the sacrum and the pelvic girdle is

A. a ball and socket joint **B.** a pivot joint
C. immovable **D.** a hinge joint

141. The joints between the centra of vertebrae are

A. freely movable **B.** slightly movable
C. immovable **D.** ball and socket joints

142. The joints between the carpals of the wrist are

A. ball and socket joints **B.** gliding joints
C. pivot joints **D.** hinge joints

143. The joint between the atlas and the axis is a

A. gliding joint **B.** fixed joint
C. hinge joint **D.** pivot joint

144. The elbow is a hinge joint formed between the humerus and the

A. scapula **B.** clavicle
C. ulna **D.** radius

145. Movement in all planes is possible at the

A. knee joint **B.** wrist joint
C. hip joint **D.** elbow joint

146. Movement in one plane only is permitted at the

A. hip joint **B.** knee joint
C. shoulder joint **D.** joint between the atlas and axis

147. The substance which lubricates bones at a movable joint is called

A. sebum **B.** endolymph
C. synovial fluid **D.** lymph

148. Friction-free movement at the knee joint is enhanced by the presence of

A. tendons **B.** synovial fluid
C. ligaments **D.** muscles

149. Which of the following tissues limits the extent to which bones can move at a joint?

A. Synovial fluid **B.** Ligaments
C. Tendons **D.** Cartilage

150. If the synovial capsule and synovial fluid of a movable joint are removed, the greatest effect on the joint would be

A. growth of the bones **B.** production of more cartilage
C. degeneration of the bone surfaces **D.** increased mobility

151. The shoulder joint is formed between the humerus and the

A. ulna **B.** scapula
C. sternum **D.** clavicle

152. The shoulder is dislocated more easily than the hip because

A. the humerus has no head
B. the acetabulum is shallow
C. the shoulder has no ligaments
D. the glenoid cavity is shallow and its ligaments are loose

153. Wearing away of the articular cartilage, which leads to painful and stiff joints, is called

A. osteoporosis **B.** rickets
C. cramp **D.** arthritis

154. Which of the following is NOT a synovial joint?

A. The hip **B.** The elbow
C. The wrist **D.** The sutures of the cranium

155. The joint between the skull and the atlas is a

A. pivot joint **B.** hinge joint
C. ball and socket joint **D.** gliding joint

SECTION H: MUSCLES AND MOVEMENT

156. The muscle which is under the control of the will is

A. skeletal muscle **B.** cardiac muscle
C. smooth muscle **D.** visceral muscle

157. Which type of muscle is involuntary but striated?

A. Voluntary muscle **B.** Skeletal muscle
C. Cardiac muscle **D.** Smooth muscle

158. Intercalated discs are found in

A. smooth muscle **B.** skeletal muscle
C. cardiac muscle **D.** tendons

159. Smooth muscle fibres are

A. branched and joined by intercalated discs
B. spindle-shaped, each with one central nucleus
C. long, cylindrical and multinucleate
D. striated and voluntary

160. Peristalsis in the alimentary canal is brought about by

A. voluntary muscle **B.** smooth muscle
C. skeletal muscle **D.** cardiac muscle

161. The muscle which contracts rhythmically throughout life without getting tired is

A. the biceps **B.** skeletal muscle
C. the intercostal muscle **D.** cardiac muscle

162. Each skeletal muscle fibre is

A. spindle-shaped **B.** unstriated
C. multinucleate **D.** branched with one nucleus

163. Muscles are attached to bones by

A. ligaments **B.** tendons
C. cartilage **D.** synovial membranes

164. Tendons are

A. made mainly of yellow elastic fibres
B. inelastic and made mainly of collagen fibres
C. elastic and join bone to bone
D. made of hyaline cartilage

165. Two bones in the mammalian body are joined by means of

A. ligaments **B.** tendons
C. muscles **D.** cartilage

166. The point of attachment of a muscle to a movable bone is the

A. insertion **B.** tendon
C. fulcrum **D.** origin

167. The biceps muscle has its origins on the

A. scapula **B.** radius
C. ulna **D.** femur

168. The biceps muscle is inserted on the

A. humerus **B.** scapula
C. radius **D.** ulna

169. The triceps muscle is inserted on the

A. radius
B. ulna, at the olecranon process
C. humerus
D. scapula

170. The triceps muscle has

A. three origins
B. one origin
C. two origins
D. four origins

171. What happens when the biceps muscle contracts? The

A. triceps muscle contracts
B. scapula moves towards the sternum
C. forearm straightens
D. forearm bends

172. Which of the following happens when a person lifts a cup of tea from a table?

A. The triceps contracts and the biceps relaxes
B. Both the biceps and triceps relax
C. The biceps contracts and the triceps relaxes
D. Both the biceps and triceps contract

173. Straightening of the arm is brought about by

A. contraction of the biceps and relaxation of the triceps
B. relaxation of both the biceps and triceps
C. contraction of both the biceps and triceps
D. contraction of the triceps and relaxation of the biceps

174. A muscle which contracts to straighten a limb is called

A. a sphincter
B. an extensor
C. a flexor
D. a ligament

175. Skeletal muscles work in antagonistic pairs because a muscle

A. is attached by ligaments
B. cannot relax
C. has two insertions
D. can only pull and cannot push

176. In the movement of the forearm, the elbow joint serves as the

A. effort
B. origin
C. fulcrum
D. load

177. Muscles act in opposite directions to each other in order to

A. regulate bodily activities
B. cause a bone to move
C. prevent muscle fatigue
D. prevent dislocation at joints

178. The energy used directly for muscle contraction is supplied by

A. ATP
B. glucose
C. lactic acid
D. glycogen

179. The condition in which a pair of antagonistic muscles contract at the same time, making it difficult to move the limb, is

A. muscle pull
B. sprain
C. cramp
D. dislocation

SECTION I: STRUCTURE OF MUSCLE AND THE SLIDING FILAMENT MODEL

180. The functional contracting unit of a myofibril is the

A. sarcomere
B. Z line
C. myosin filament
D. muscle fibre

181. The cytoplasm of a muscle fibre is called the

A. myofibril **B.** sarcolemma
C. sarcoplasmic reticulum **D.** sarcoplasm

182. The thin filaments of a myofibril are made of the protein

A. myosin **B.** actin
C. keratin **D.** collagen

183. Calcium ions needed for muscle contraction are released from the

A. sarcolemma **B.** mitochondria
C. nucleus **D.** sarcoplasmic reticulum

184. In a resting muscle, the binding sites on the actin filaments are covered by

A. myosin heads **B.** ATP
C. calcium ions **D.** tropomyosin

185. During the contraction cycle, the myosin heads detach from the actin filaments when

A. tropomyosin is removed **B.** ATP binds to them
C. calcium ions are released **D.** acetylcholine is released

186. According to the sliding filament model, a muscle shortens because

A. both actin and myosin filaments shorten
B. the Z lines move further apart
C. actin filaments slide over myosin filaments towards the centre of the sarcomere
D. only the myosin filaments shorten

SECTION J: FUNCTIONS OF THE SKELETON, INJURIES AND DISORDERS

187. Red blood cells are produced in the

A. red bone marrow **B.** synovial membrane
C. yellow bone marrow **D.** cartilage

188. A fracture in which the broken bone pierces through the skin is a

A. simple fracture **B.** compound fracture
C. greenstick fracture **D.** comminuted fracture

189. Greenstick fractures are most common in children because their bones

A. are still flexible **B.** contain more calcium
C. are fully ossified **D.** lack collagen

190. Over-stretching or tearing of the ligaments at a joint is called a

A. strain **B.** fracture
C. dislocation **D.** sprain

191. Rickets in children is caused by a deficiency of

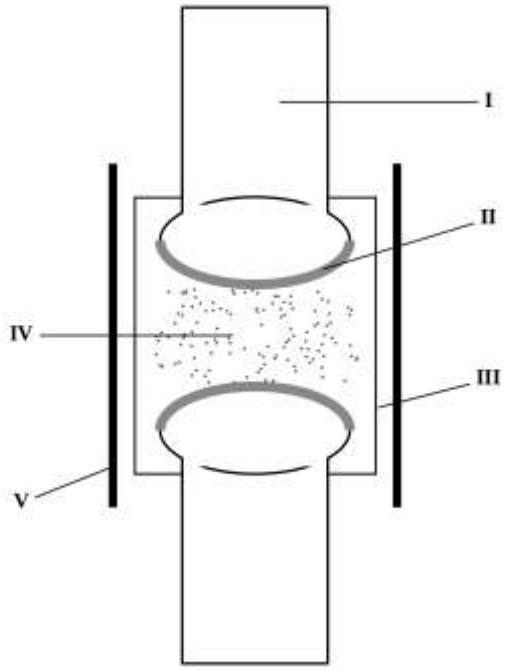
A. vitamin K **B.** vitamin A
C. vitamin C **D.** vitamin D

192. The first aid for a person with a suspected fractured arm is to

A. massage the arm firmly
B. push the bone back into place
C. ask the person to move the arm
D. immobilise the arm with a splint or sling

SECTION K: DIAGRAM-BASED QUESTIONS

Study the diagram below of a typical movable (synovial) joint and use it to answer Questions 193 to 196.



193. The part labelled II is the

A. tendon
B. ligament
C. synovial membrane
D. articular cartilage

194. The function of the part labelled III is to

A. join bone to bone
B. join muscle to bone
C. secrete synovial fluid
D. produce blood cells

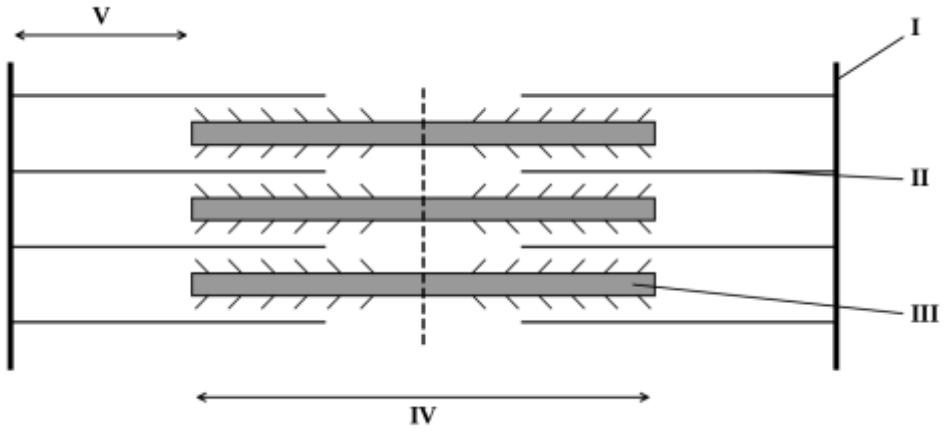
195. The part labelled V, which holds the bones together, is the

A. synovial membrane
B. ligament
C. articular cartilage
D. tendon

196. The substance in the part labelled IV

A. joins muscle to bone
B. lubricates the joint and reduces friction
C. stores calcium
D. produces red blood cells

The diagram below shows a sarcomere of a skeletal muscle. Use it to answer Questions 197 to 200.



197. The line labelled I is the

A. A band
B. H zone
C. M line
D. Z line

198. The filament labelled II is made of

A. keratin
B. collagen
C. myosin
D. actin

199. During contraction, the region labelled IV

A. remains the same length
B. shortens
C. lengthens
D. disappears

200. The region labelled V represents

A. the part of actin that does not overlap with myosin
B. the region where actin and myosin overlap
C. the line that joins the myosin filaments
D. the whole length of the myosin filaments

ANSWER SECTION

(Detach before issuing to students if required)

Quick Answer Grid

1. C	**21.** B	**41.** D	**61.** A	**81.** A	**101.** C	**121.** B	**141.** B	**161.** D	**181.** D
2. A	**22.** C	**42.** A	**62.** A	**82.** D	**102.** A	**122.** D	**142.** B	**162.** C	**182.** B
3. C	**23.** D	**43.** B	**63.** B	**83.** C	**103.** D	**123.** D	**143.** D	**163.** B	**183.** D
4. B	**24.** B	**44.** C	**64.** C	**84.** C	**104.** B	**124.** C	**144.** C	**164.** B	**184.** D
5. A	**25.** D	**45.** A	**65.** A	**85.** C	**105.** C	**125.** D	**145.** C	**165.** A	**185.** B
6. D	**26.** C	**46.** C	**66.** B	**86.** B	**106.** D	**126.** D	**146.** B	**166.** A	**186.** C
7. B	**27.** B	**47.** B	**67.** C	**87.** C	**107.** C	**127.** A	**147.** C	**167.** A	**187.** A
8. B	**28.** C	**48.** A	**68.** A	**88.** A	**108.** A	**128.** D	**148.** B	**168.** C	**188.** B
9. C	**29.** B	**49.** A	**69.** D	**89.** A	**109.** B	**129.** C	**149.** B	**169.** B	**189.** A
10. B	**30.** B	**50.** D	**70.** D	**90.** A	**110.** B	**130.** C	**150.** C	**170.** A	**190.** D
11. A	**31.** C	**51.** D	**71.** D	**91.** B	**111.** B	**131.** C	**151.** B	**171.** D	**191.** D
12. A	**32.** C	**52.** B	**72.** D	**92.** A	**112.** D	**132.** D	**152.** D	**172.** C	**192.** D
13. D	**33.** D	**53.** A	**73.** D	**93.** D	**113.** A	**133.** D	**153.** D	**173.** D	**193.** D
14. A	**34.** A	**54.** C	**74.** C	**94.** B	**114.** D	**134.** C	**154.** D	**174.** B	**194.** C
15. C	**35.** C	**55.** B	**75.** B	**95.** C	**115.** B	**135.** A	**155.** B	**175.** D	**195.** B
16. B	**36.** B	**56.** B	**76.** A	**96.** B	**116.** A	**136.** D	**156.** A	**176.** C	**196.** B
17. D	**37.** C	**57.** B	**77.** B	**97.** D	**117.** B	**137.** A	**157.** C	**177.** B	**197.** D
18. C	**38.** C	**58.** B	**78.** C	**98.** C	**118.** A	**138.** D	**158.** C	**178.** A	**198.** D
19. A	**39.** A	**59.** C	**79.** C	**99.** C	**119.** A	**139.** C	**159.** B	**179.** A	**199.** A
20. D	**40.** A	**60.** C	**80.** A	**100.** D	**120.** B	**140.** C	**160.** B	**180.** A	**200.** A

Answer distribution — A: 46 B: 53 C: 52 D: 49

Answer Key with Correct Options

No.	Ans.	Correct option	No.	Ans.	Correct option
1	C	Locomotion	101	C	A nerve, an artery and a vein
2	A	Skeletal, muscular and nervous systems	102	A	Sigmoid notch of the ulna
3	C	Hydrostatic	103	D	Olecranon process
4	B	Chitin	104	B	Ulna
5	A	Calcium salts	105	C	Coronoid fossa
6	D	Calcium carbonate	106	D	Ulna
7	B	Shark	107	C	Radius
8	B	Cannot grow with the animal	108	A	Carpals
9	C	It prevents desiccation	109	B	Femur
10	B	Soon after moulting	110	B	Acetabulum
11	A	Grows with the body and is relatively light	111	B	Trochanters
12	A	Shorter and fatter	112	D	Condyles
13	D	Chaetae	113	A	Leg from bending forwards at the knee
14	A	Coelom	114	D	Tibia and fibula
15	C	Cockroach	115	B	Calcaneum
16	B	Water is needed to maintain the skeletal fluid	116	A	4
17	D	Muscles are attached to its outer surface	117	B	2 phalanges
18	C	Digestion of food	118	A	Humerus and femur
19	A	Becomes rigid when its spongy tissue fills with blood under pressure	119	A	Pentadactyl limbs
20	D	Chitin, bone and cartilage	120	B	Limb girdles
21	B	Connective tissues	121	B	Scapula, coracoid and clavicle
22	C	Calcium phosphate	122	D	Scapula
23	D	Collagen	123	D	Clavicle
24	B	Hydroxyapatite	124	C	Scapula
25	D	Ossification	125	D	Spine
26	C	Harder but more brittle	126	D	Acromion process
27	B	Osteoblasts	127	A	Muscles
28	C	Osteocytes	128	D	Allows a wide range of movement of the fore limb
29	B	Osteoclasts	129	C	Innominate bone
30	B	Their bones are more brittle and lose minerals	130	C	Ilium, ischium and pubis
31	C	Haversian system	131	C	Ilium
32	C	Haversian canal	132	D	Pubic symphysis
33	D	Lamellae	133	D	Obturator foramen
34	A	Lacunae	134	C	Allows the passage of nerves and blood vessels
35	C	Canaliculi	135	A	Pelvic girdle and the sacrum

No.	Ans.	Correct option	No.	Ans.	Correct option
36	B	Chondrin	136	D	Bear the weight of the body without dislocating
37	C	Chondrocytes	137	A	The pelvic girdle is fused to the sacrum but the pectoral girdle is not
38	C	Has no blood vessels	138	D	Forms the birth canal
39	A	It is avascular	139	C	Joint
40	A	Fibrocartilage	140	C	Immovable
41	D	Hyaline cartilage	141	B	Slightly movable
42	A	Elastic cartilage	142	B	Gliding joints
43	B	Both have cells in lacunae	143	D	Pivot joint
44	C	Haversian canals	144	C	Ulna
45	A	Hyaline cartilage	145	C	Hip joint
46	C	Skull, vertebral column, ribs and sternum	146	B	Knee joint
47	B	Atlas	147	C	Synovial fluid
48	A	Limbs and limb girdles	148	B	Synovial fluid
49	A	Lower jaw	149	B	Ligaments
50	D	Sutures	150	C	Degeneration of the bone surfaces
51	D	Foramen magnum	151	B	Scapula
52	B	Atlas	152	D	The glenoid cavity is shallow and its ligaments are loose
53	A	Protect the brain	153	D	Arthritis
54	C	Absorb shock and prevent friction between vertebrae	154	D	The sutures of the cranium
55	B	Ligaments	155	B	Hinge joint
56	B	7	156	A	Skeletal muscle
57	B	33	157	C	Cardiac muscle
58	B	Five sacral vertebrae	158	C	Cardiac muscle
59	C	Caudal vertebrae	159	B	Spindle-shaped, each with one central nucleus
60	C	Vertebrarterial canals	160	B	Smooth muscle
61	A	Centrum	161	D	Cardiac muscle
62	A	Spinal cord	162	C	Multinucleate
63	B	Surfaces for muscle attachment	163	B	Tendons
64	C	Prezygapophyses	164	B	Inelastic and made mainly of collagen fibres
65	A	Gliding joints	165	A	Ligaments
66	B	Vertebral artery	166	A	Insertion
67	C	Cervical vertebra	167	A	Scapula
68	A	Cervical vertebrae	168	C	Radius
69	D	Atlas	169	B	Ulna, at the olecranon process
70	D	Skull and the axis	170	A	Three origins
71	D	Centrum	171	D	Forearm bends

No.	Ans.	Correct option	No.	Ans.	Correct option
72	D	Nodding of the head	172	C	The biceps contracts and the triceps relaxes
73	D	Axis vertebra	173	D	Contraction of the triceps and relaxation of the biceps
74	C	Provides a point for pivoting	174	B	An extensor
75	B	Prezygapophyses	175	D	Can only pull and cannot push
76	A	Absent centrum and wing-like transverse processes	176	C	Fulcrum
77	B	A thoracic vertebra	177	B	Cause a bone to move
78	C	Demifacets on the centra of thoracic vertebrae	178	A	ATP
79	C	Facet on the transverse process of a thoracic vertebra	179	A	Muscle pull
80	A	Lumbar vertebra	180	A	Sarcomere
81	A	Bear the weight of the upper body	181	D	Sarcoplasm
82	D	Lumbar vertebra	182	B	Actin
83	C	Long and point forwards and downwards	183	D	Sarcoplasmic reticulum
84	C	Pelvic girdle	184	D	Tropomyosin
85	C	Sacral wings	185	B	ATP binds to them
86	B	Cervical, thoracic, lumbar, sacral, caudal	186	C	Actin filaments slide over myosin filaments towards the centre of the sarcomere
87	C	12	187	A	Red bone marrow
88	A	True ribs	188	B	Compound fracture
89	A	2	189	A	Are still flexible
90	A	Floating ribs	190	D	Sprain
91	B	False ribs	191	D	Vitamin D
92	A	Manubrium	192	D	Immobilise the arm with a splint or sling
93	D	Costal cartilage	193	D	Articular cartilage
94	B	Intercostal muscles	194	C	Secrete synovial fluid
95	C	Pentadactyl	195	B	Ligament
96	B	Humerus	196	B	Lubricates the joint and reduces friction
97	D	Radius and ulna	197	D	Z line
98	C	Glenoid cavity	198	D	Actin
99	C	Tendon of the biceps	199	A	Remains the same length
100	D	Attachment of the deltoid muscle	200	A	The part of actin that does not overlap with myosin

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